

Critical Analysis: Large Language Models are Zero-Shot Reasoners

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1 Methodological Strengths

1.1 Broad and structured experimental design

A primary methodological strength lies in the breadth and structure of the experimental evaluation. The paper evaluates Zero-shot Chain-of-Thought prompting (Zero-shot-CoT) on twelve datasets that together cover arithmetic reasoning, commonsense reasoning, symbolic reasoning and other logical reasoning tasks. The suite includes:

- Arithmetic reasoning: SingleEq, AddSub, MultiArith, GSM8K, AQUA-RAT, SVAMP.
- Commonsense reasoning: CommonsenseQA, StrategyQA.
- Symbolic reasoning: Last Letters, Coin Flip.
- Other logical reasoning: Date Understanding, Tracking Shuffled Objects.

The datasets vary in answer format, including numeric answers, multiple-choice answers, yes-or-no answers and short free-form strings. Across all these tasks, the authors adopt a single task-agnostic Zero-shot-CoT template rather than designing separate prompts per dataset. This choice strengthens the claim that the method targets general reasoning behaviour instead of carefully tuned task-specific heuristics.

The experimental protocol is also clearly separated into task categories. Arithmetic datasets test multi-step numerical computation, symbolic datasets test deterministic state tracking or string manipulation, and BIG-bench style tasks test temporal and object-state reasoning. This organisation makes it easier to interpret where the method succeeds and where it does not.

1.2 Strong baselines and layered comparisons

Another important strength is the careful choice of baselines. The paper does not compare Zero-shot-CoT only against weak zero-shot prompts, but rather against a hierarchy of increasingly powerful methods:

- Standard zero-shot prompting with a simple answer cue.
- Standard few-shot prompting with short answers.
- Few-shot Chain-of-Thought prompting (Few-shot-CoT) that uses step-by-step rationales as demonstrations.
- A fine-tuned GPT-3 model with a verifier on GSM8K.

- PaLM models evaluated with and without Zero-shot-CoT and self-consistency decoding.

For MultiArith with text-davinci-002, standard zero-shot prompting achieves 17.7 percent accuracy, standard few-shot prompting with eight examples yields 33.8 percent, Zero-shot-CoT reaches 78.7 percent, and Few-shot-CoT with eight examples further improves to 93.0 percent. For GSM8K with the same model, standard zero-shot prompting attains 10.4 percent, standard few-shot prompting reaches 15.6 percent, Zero-shot-CoT reaches 40.7 percent, Few-shot-CoT reaches 48.7 percent, and a mixed prompt combining Zero-shot-CoT and Few-shot-CoT achieves about 51.5 percent, compared with around 55 percent for a fine-tuned 175B GPT-3 plus verifier system. For PaLM 540B, Zero-shot-CoT improves GSM8K accuracy from 12.5 percent to 43.0 percent, Zero-shot-CoT with self-consistency further raises accuracy to 70.1 percent, and Few-shot-CoT with self-consistency reaches 74.4 percent.

These layered comparisons support nuanced conclusions about when prompt-only methods can approach or surpass fine-tuned systems.

1.3 Explicit prompting pipeline and answer extraction

The methodology for Zero-shot-CoT is described as a clear two-stage prompting pipeline.

Stage 1: reasoning extraction. For a question x , the method constructs a prompt of the form

Q: $[x]$. A: Let us think step-by-step.

The model is decoded with greedy decoding, producing a chain-of-thought text z . This text explicitly contains intermediate reasoning steps such as restating quantities, computing partial results and gradually arriving at a candidate answer.

Stage 2: answer extraction. The second stage builds a new prompt by concatenating the Stage 1 prompt, the generated reasoning z , and an answer cue tailored to the answer format. For numeric tasks, the cue is

Therefore, the answer (arabic numerals) is

For multiple-choice tasks, the cue is of the form

Therefore, among A through E, the answer is

The model is called again with greedy decoding, and the resulting continuation is parsed to extract a single answer. The answer-cleansing procedure is simple and deterministic: the system scans from left to right for the first digit sequence, the first valid option label, or the first occurrence of “Yes” or “No”, depending on the task. The same parsing rules are applied to all methods, which reduces evaluation bias.

The explicit separation between reasoning generation and answer extraction is methodologically valuable because it clarifies how chain-of-thought text is used and avoids entangling reasoning with answer formatting.

1.4 Scaling and prompt-robustness analyses

The paper conducts a detailed scaling analysis across model sizes and families. For arithmetic benchmarks such as MultiArith and GSM8K, standard zero-shot prompting yields performance curves that remain almost flat as model size increases. In contrast, Zero-shot-CoT produces accuracy that grows significantly with scale.

On MultiArith, the Zero-shot-CoT accuracy for original GPT-3 models (0.3B, 1.3B, 6.7B, 175B) is 1.7, 2.2, 2.3 and 19.0 percent respectively, which shows only modest gains from scaling. For InstructGPT variants (S, M, L, XL-1, XL-2 corresponding to text-ada-001, text-babbage-001, text-curie-001, text-davinci-001 and text-davinci-002), Zero-shot-CoT accuracy rises from 2.0 percent at the smallest model to 3.7, 3.3 and 47.8 percent for medium to XL-1, and finally to 78.7 percent for text-davinci-002. Combined with the fact that text-davinci-002 improves from 17.7 percent under standard zero-shot prompting to 78.7 percent under Zero-shot-CoT, these results indicate that substantial reasoning gains emerge only when both scale and instruction tuning are present.

For PaLM models on GSM8K, Table 27 reports that standard zero-shot prompting achieves 2.1, 7.0 and 12.5 percent accuracy for 8B, 62B and 540B models, while Zero-shot-CoT improves these numbers to 2.4, 10.5 and 43.0 percent. The corresponding PaLM 540B experiments with self-consistency show that sampling multiple reasoning paths and aggregating answers further increases accuracy.

The prompt-robustness study explores alternative trigger phrases for MultiArith. Variants such as “First,” or “Let us think about this logically.” lead to accuracy above 70 percent, whereas misleading instructions such as “Do not think. Just feel.” or irrelevant sentences do not improve over the 17.7 percent zero-shot baseline. The default phrase “Let us think step-by-step.” remains the strongest at 78.7 percent. This experiment provides evidence that the gains are driven by the semantic intent to encourage step-by-step reasoning rather than arbitrary prompt perturbations.

1.5 Qualitative error analysis

The inclusion of qualitative examples is another methodological advantage. The paper shows CommonsenseQA instances where Zero-shot-CoT generates logically reasonable chains-of-thought but ultimately selects the wrong option, as well as examples where the model produces multiple plausible choices instead of committing to one. For arithmetic tasks such as MultiArith, the analysis distinguishes between Zero-shot-CoT errors where an initially correct intermediate result is corrupted by extra steps and Few-shot-CoT errors where the chain-of-thought misapplies arithmetic operations. These observations help to characterise how different prompting strategies fail and support later discussions in the paper.

2 Key Limitations

2.1 Benchmark and task coverage

Despite the relatively wide set of benchmarks, several coverage limitations remain:

- All tasks are short, static, text-only prompts and single-turn question answering. The paper does not consider multi-turn dialogue, interactive problem solving or long-context reasoning involving entire documents.
- The benchmarks are predominantly synthetic or curated datasets. Real-world settings such as scientific question answering, legal reasoning, or domain-specific decision support are absent, so external validity beyond benchmark-style problems is not directly established.
- Commonsense reasoning is represented by only two datasets, and the results are mixed: Zero-shot-CoT improves StrategyQA but slightly degrades performance on CommonsenseQA. The paper does not extend this comparison to a broader spectrum of commonsense benchmarks, which would help to clarify when chain-of-thought prompting benefits commonsense tasks.

These limitations imply that the claim that large language models are “zero-shot reasoners” is most strongly supported for short-form reasoning benchmarks, but less clearly substantiated for complex real-world reasoning scenarios.

2.2 Closed-source models and limited training transparency

The methodology relies on proprietary LLMs such as GPT-3, InstructGPT and PaLM, accessed through APIs or internal infrastructure. The architecture, pretraining data distribution and instruction-tuning protocols are controlled by external sources and are not fully transparent. This leads to several issues:

- Improvements from Zero-shot-CoT may depend on particular properties of InstructGPT or PaLM, such as the degree of instruction tuning, but the paper cannot fully disentangle these effects.
- It is difficult to generalise the findings to open-source models or future architectures without similar pretraining regimes.
- The analysis cannot examine how training data explicitly shapes chain-of-thought behaviour, for example whether models have seen step-by-step mathematical explanations during pretraining.

These constraints limit the ability to draw principled conclusions about architecture-level requirements for Zero-shot-CoT success.

2.3 Narrow metrics and missing calibration or faithfulness evaluation

Accuracy is used as the sole quantitative metric for all tasks. While this is standard for many benchmarks, it leaves several aspects of model behaviour unmeasured:

- There is no evaluation of partial correctness or reasoning quality. For instance, a chain-of-thought may correctly derive key intermediate results but fail at the final step; such cases are treated as entirely wrong.
- Calibration and confidence are not reported. There is no analysis of whether Zero-shot-CoT improves or worsens the alignment between model confidence and correctness.
- Faithfulness of explanations is not assessed. The paper does not test whether the chains-of-thought reflect the internal computation of the model or are post-hoc rationalisations.

Moreover, the paper does not provide confidence intervals or variance estimates for accuracy, particularly for few-shot baselines where seed and example ordering can influence outcomes. This omission makes it harder to quantify the statistical robustness of the reported gains.

2.4 Computational cost and latency considerations

Zero-shot-CoT inherently consumes more computation than standard zero-shot prompting because each query requires at least two full forward passes: one for reasoning and one for the final answer. When combined with self-consistency for PaLM 540B, multiple chains-of-thought are sampled and aggregated, further increasing inference cost. The paper does not quantify:

- The relative latency or computational overhead of Zero-shot-CoT compared with standard prompting and Few-shot-CoT.

- The trade-off between the number of sampled chains-of-thought and achievable accuracy when using self-consistency decoding.

Without such analysis, it is difficult to judge whether Zero-shot-CoT is suitable for deployment in resource-constrained environments.

2.5 Partial coverage of failure modes

The error analysis presented in the paper is insightful but limited in scope:

- For CommonsenseQA, the paper highlights that Zero-shot-CoT can produce reasonable explanations but incorrect answers. However, it does not categorise which question types are most prone to such discrepancies.
- For symbolic tasks such as Last Letters and Coin Flip, where Zero-shot-CoT improves accuracy dramatically, the study does not explore performance under increased sequence length or more complex state transitions.
- Prompt robustness is analysed on MultiArith, but there is no discussion of adversarial prompts or perturbations designed to mislead chain-of-thought generation.

These gaps restrict the understanding of when Zero-shot-CoT fails and how fragile it is to distribution shifts or adversarial input design.

3 Technical Bottlenecks

3.1 Dependence on model scale

The scaling experiments reveal a structural bottleneck: Zero-shot-CoT only provides substantial benefits when applied to sufficiently large models. For smaller GPT-3 variants and other models such as GPT-2, GPT-Neo, GPT-J, T0 and OPT, Zero-shot-CoT does not improve and sometimes even reduces accuracy compared with standard zero-shot prompting on MultiArith and GSM8K. Significant gains appear only for large InstructGPT models and large PaLM models.

This dependence on scale indicates that the chain-of-thought trigger is exploiting internal computations that emerge only after extensive pretraining on large models. In environments where such models are too costly or cannot be deployed, Zero-shot-CoT alone cannot compensate for the lack of capacity.

3.2 Two-stage prompting pipeline constraints

The two-stage pipeline introduces several technical constraints:

- It doubles the number of forward passes per query compared with standard prompting, which directly increases latency and cost.
- It relies on the model’s ability to condition on its own chain-of-thought text without overfitting to verbal noise. Errors in Stage 1 can propagate and become amplified in Stage 2.
- It consumes context length, since the full question, trigger, reasoning and answer cue must all fit within the model’s maximum sequence length. For longer inputs or more verbose chains-of-thought, this may force truncation.

These factors constitute an architectural bottleneck that limits the method’s scalability to long-context or high-throughput settings.

3.3 Prompt and answer-format brittleness

Zero-shot-CoT exhibits sensitivity to prompt semantics and answer-format design:

- The trigger phrase must convey an instruction to reason step-by-step. Misleading or unrelated phrases do not yield gains, and some instructive variants perform worse than the default.
- The answer-extraction cues are tailored to specific formats, such as numeric or multiple-choice answers. If these cues are misaligned with a dataset’s structure or a model’s learned patterns, accuracy can degrade.
- The cross-task Few-shot-CoT experiments suggest that in-context learning is heavily influenced by surface-level answer patterns. Similar phenomena are likely to affect Zero-shot-CoT when triggers or answer cues inadvertently encode unintended biases.

This brittleness reveals an information bottleneck: models are sensitive to shallow prompt structures, and reasoning performance is not purely determined by deep task semantics.

3.4 Lack of explicit verification or symbolic checking

Although Zero-shot-CoT yields explicit chains-of-thought, the method does not incorporate any explicit verification or checking mechanism:

- Arithmetic operations inside the chain-of-thought are not validated, even when they violate basic numerical constraints.
- For symbolic tasks, such as Coin Flip and Last Letters, there is no external check that state transitions or string manipulations follow the problem rules.
- Self-consistency uses majority voting over answers but ignores disagreement at the level of reasoning paths.

As a result, the approach can produce fluent but incorrect explanations, limiting its reliability in safety-critical applications.

4 Research Implications

4.1 Latent zero-shot reasoning in large models

The key empirical result that a single sentence such as “Let us think step-by-step.” can lift MultiArith accuracy from 17.7 percent to 78.7 percent and GSM8K accuracy from 10.4 percent to 40.7 percent with the same model and training data indicates that large LLMs already encode substantial multi-step reasoning capabilities that are not revealed by naive prompts. This supports the view that prompting acts as a control interface that can expose or suppress latent abilities.

Such evidence motivates more systematic study of how prompting strategies interact with pretraining, and how much of reasoning behaviour is already present in pre-trained models versus created by fine-tuning or additional supervision.

4.2 Prompting as a mechanism for controlling system-1 and system-2 modes

The contrast between standard zero-shot prompting and Zero-shot-CoT suggests that LLMs can operate in different regimes. Direct answer prompts elicit fast, heuristic responses resembling system-1 behaviour, whereas chain-of-thought triggers induce slower, multi-step reasoning more similar to system-2 processing. The same underlying model can switch regimes based on a few words in the prompt.

This observation has practical implications for the design of interfaces and APIs. It implies that reasoning mode can be controlled without architectural changes or fine-tuning, which opens directions for user-facing systems that adapt prompting strategies dynamically depending on task difficulty or safety requirements.

4.3 Reconsidering the role of few-shot examples

The paper demonstrates that strong reasoning performance is not solely a product of few-shot learning. Although Few-shot-CoT remains superior on some benchmarks, Zero-shot-CoT narrows the gap substantially without any task-specific examples, and Zero-shot-CoT plus self-consistency with PaLM 540B achieves 70.1 percent accuracy on GSM8K, surpassing several previously reported fine-tuned baselines.

This suggests that in-context examples complement but do not fully define reasoning behaviour. Task-agnostic prompts can already elicit substantial reasoning capability, and few-shot exemplars primarily refine and stabilise that behaviour. This insight can shape future benchmark protocols and prevent over-attributing performance gains to example-based learning alone.

4.4 Implications for benchmark and evaluation design

The findings also affect how reasoning benchmarks should be read and designed:

- Evaluations that use only direct answer prompts underestimate what pre-trained models can achieve with more sophisticated zero-shot prompting.
- Benchmark reports should specify the prompting protocol in detail, including whether chain-of-thought triggers and multi-stage pipelines are used, to ensure fair comparison.
- The fact that symbolic tasks such as Last Letters and Coin Flip show dramatic gains under Zero-shot-CoT suggests that some benchmarks are as much tests of prompt semantics as of underlying reasoning, and that new benchmarks may be needed to isolate reasoning from prompt engineering.

5 Potential Research Directions

5.1 Automatic discovery and optimisation of reasoning prompts

One concrete direction is to move beyond manually crafted phrases and develop algorithms that automatically search for effective reasoning prompts. Possible approaches include:

- Gradient-free optimisation in discrete prompt space to maximise validation accuracy under chain-of-thought generation.
- Meta-learning frameworks that learn families of prompts across many tasks, with different prompts corresponding to different reasoning styles.
- Reinforcement learning where the reward signal reflects both accuracy and stability of reasoning across perturbations.

Such techniques could generalise the idea of Zero-shot-CoT and reduce dependence on specific hand-written triggers.

5.2 Verifier-guided reasoning and calibration

Another direction is to integrate Zero-shot-CoT with explicit checking mechanisms:

- Training or prompting separate verifier models to score chain-of-thought outputs for logical coherence and correctness, and to filter out low-quality reasoning before answer extraction.
- Incorporating lightweight symbolic solvers for arithmetic and state-tracking subproblems, using the chain-of-thought as a program sketch that is then executed and verified.
- Designing self-consistency schemes that aggregate reasoning paths based not only on answer majority but also on cross-path agreement and internal consistency.

These additions could improve both accuracy and calibration, and help distinguish genuinely well-reasoned answers from plausible but incorrect ones.

5.3 Architectural support for explicit intermediate computation

The observed benefits of chain-of-thought prompting suggest that explicit intermediate computation is helpful. Future model designs could support this more directly by:

- Incorporating scratchpad mechanisms or memory modules whose content is trained to align with chain-of-thought outputs.
- Using architectures that segment computation into discrete reasoning steps, with attention constrained across steps to enforce structured progression.
- Introducing objectives that explicitly reward consistency between internal latent trajectories and externalised chains-of-thought.

Such architectures might reduce the reliance on very large parameter counts and make structured reasoning more robust.

5.4 Expanded benchmarks and multi-dimensional evaluation

The field would benefit from benchmarks that are explicitly designed with chain-of-thought prompting in mind. These benchmarks could:

- Include tasks that require revising or correcting earlier reasoning, which would test whether chain-of-thought generation can handle backtracking and self-correction.
- Measure explanation quality and faithfulness through human evaluation or automatic proxies, rather than only final answer accuracy.
- Incorporate longer contexts, multi-turn interactions and realistic domain constraints to test scalability of the two-stage pipeline.

In addition, new metrics for explanation faithfulness, calibration and robustness under input perturbations would complement accuracy and provide a more complete picture of reasoning performance.

5.5 Extensions to multimodal and interactive settings

The Zero-shot-CoT idea can also be explored in settings beyond text-only, single-turn question answering:

- Applying step-by-step reasoning prompts to multimodal models that process images, video or structured data, and evaluating their effect on visual question answering and grounded reasoning tasks.
- Using chain-of-thought-like triggers in reinforcement learning or planning contexts, where the model is asked to verbalise its intended action sequence before executing it.
- Designing interactive systems that dynamically decide when to invoke chain-of-thought reasoning based on uncertainty estimates or task criticality, balancing accuracy with cost.

These directions would test whether the gains observed in text benchmarks generalise to more complex, interactive environments.

6 Conclusion

The paper “Large Language Models are Zero-Shot Reasoners” provides strong evidence that large language models possess substantial latent multi-step reasoning capabilities that can be elicited by a simple, task-agnostic chain-of-thought trigger. Methodologically, the work is distinguished by its broad and structured experiment design, strong baselines, explicit two-stage prompting pipeline, scaling analysis and targeted prompt robustness study. The central result is that a fixed phrase such as “Let us think step-by-step.” can dramatically increase zero-shot accuracy on challenging arithmetic and symbolic reasoning benchmarks, and that combining Zero-shot-CoT with self-consistency can match or exceed fine-tuned systems on GSM8K with PaLM 540B.

At the same time, the analysis reveals key limitations, including dependence on very large models, coverage restricted to short-form text benchmarks, the absence of calibration or faithfulness metrics, and the lack of explicit verification mechanisms. These limitations highlight technical bottlenecks in model scale, inference cost, prompt brittleness and unverified reasoning.

The most promising research directions include automatic optimisation of reasoning prompts, integration with verifier models and symbolic solvers, architectural designs that encode explicit intermediate computation, expanded benchmarks that evaluate both answers and explanations, and extensions of chain-of-thought prompting to multimodal and interactive settings. Together, these avenues suggest a path toward more reliable and controllable zero-shot reasoning in future large language models.